

# EXPLAINABLE MEDICAL AI DIAGNOSTIC REPORT

Automated Chest X-Ray Screening for Pneumonia Detection with Visual Explanations

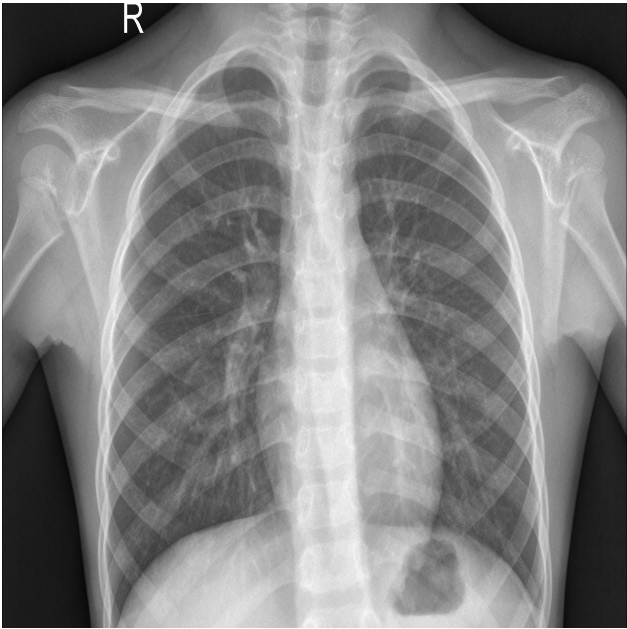
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|----------------|----------------|----------------|---------------------|
| Patient Name:  | Compare Scan A | Report Date:   | 2026-07-03 00:44:06 |
| Diagnostic ID: | REC-#8         | Model Version: | PneumoniaCNN v1.0.0 |

**AI DIAGNOSTIC RESULT: PNEUMONIA**  
Probabilities — Pneumonia: 50.55% | Normal: 49.45%

**Confidence score: 50.55%**  
Analysis Processing Time: 24.48 seconds

## IMAGE FINDINGS & EXPLAINABLE AI ANALYSIS

Original Chest X-Ray



Grad-CAM Localization Heatmap



## AI DIAGNOSTIC FINDINGS CHECKLIST

✓ Lung Opacity: The model detected areas of increased density in pulmonary fields.

✓ Lower Lobe Abnormalities: Focal densities and consolidations localized in the lobes.

✓ High Density Infiltrates: Fluid density regions detected, strongly shifting probability values.

**Visual Explanation Summary:** The AI model identified localized regions of high density and opacity, highlighted by the red/yellow regions in the Grad-CAM heatmap above. These patterns correspond to consolidated lung parenchyma and fluid infiltrates, which are standard clinical indicators of pneumonia infection.

**Disclaimer:** This diagnostic report was generated by an experimental Artificial Intelligence system. It is intended solely for educational, portfolio demonstration, and decision-support purposes. This report does NOT constitute a medical diagnosis. The findings and explanations should always be verified by a board-certified physician or radiologist.